

BANARAS HINDU UNIVERSITY

B.A. (Hons.) Semester I Examination, 2022-23

Paper No. STA-101: Statistical Methods and Probability

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) State the advantages of graphical representation of data. Explain the histogram with an example. [7]
 - (b) Differentiate between primary data and secondary data. Describe the different sources of secondary data in India. [7]
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Question 2.

- (a) Describe the different measures of central tendency. Briefly explain the meaning of an outlier. Using an example, show how an outlier can affect the value of the arithmetic mean. [7]
 - (b) Define median for ungrouped data when the number of observations is even. Derive the expression to obtain the median for a continuous frequency distribution. [7]
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Question 3.

- (a) State the relative and absolute measures of dispersion and describe the main difference between mean deviation and standard deviation. [7]
 - (b) Define the relation between moments about the mean in terms of moments about an arbitrary point, and vice versa. [7]
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Question 4.

- (a) Describe the absolute measures of skewness. Explain the coefficient of skewness based upon moments (β_1 and γ_1). [7]
- (b) Discuss the limitations of the classical definition of probability. For any two events E_1 and E_2 , show that:

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2).$$

Also, for events $E_1, E_2, \dots, E_n$, show that:

- (i) $P(E_1 \cap E_2 \cap \cdots \cap E_n) \geq P(E_1) + P(E_2) + \cdots + P(E_n) - (n - 1)$
- (ii) $P(E_1 \cup E_2 \cup \cdots \cup E_n) \leq P(E_1) + P(E_2) + \cdots + P(E_n)$

[7]

Question 5.

- (a) For any three events E_1 , E_2 and E_3 , show that:

$$P(E_1 \cup E_2 \mid E_3) = P(E_1 \mid E_3) + P(E_2 \mid E_3) - P(E_1 \cap E_2 \mid E_3).$$

[7]

- (b) In two companies A and B, a student of Statistics has applied for a job. The probability of his selection in company A is 0.7 and of his rejection in company B is 0.5. The probability of at least one of his applications being rejected is 0.6. What is the probability that he will be selected in exactly one of the companies?

[7]

Question 6.

- (a) What do you understand by the distribution function of a random variable? Describe the important properties of a distribution function.
- (b) What do you mean by a probability density function (p.d.f.)? Let X be a continuous random variable with p.d.f. given by:

$$f(x) = \begin{cases} px, & 0 < x < 1 \\ p, & 1 \leq x < 2 \\ -px + 3p, & 2 \leq x < 3 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Determine the constant p .
- (ii) Determine the cumulative distribution function (CDF).
- (iii) If three independent observations are made, what is the probability that exactly one of these three numbers is larger than 1.5?

[7]

Question 7.

- (a) Describe the conditional probability distribution function. The joint probability distribution of two random variables X and Y is given below:

$X \backslash Y$	1	2	3	4
1	$1/9$	$1/2$	$1/18$	$1/36$
2	$1/2$	$1/2$	$1/8$	$1/36$
3	$5/36$	$1/36$	$1/36$	$1/36$
4	$1/36$	$1/18$	$1/36$	$5/36$

Find the conditional distribution of X given $Y = 1$. [7]

- (b) Show that the mathematical expectation of the sum of n random variables is equal to the sum of their individual expectations, provided all expectations exist. [7]